



CIT316 – Advanced Software Design

Tutorial - 06

Part one:

1. What is LSP briefly explain your answer with real life example?
2. What is the meaning of DIP?
3. An application directly connects to a MySQLDatabase class. The company wants to switch to MongoDB later.
 - 3.1 What issue arises from this design?
 - 3.2 How can DIP help make the system flexible?
- 4 Why do we need SQA and explain SQA vs Testing?
- 5 How do you ensure high-quality software before release?

Part Two:

Scenario based questions, justify your answers.

1. **Scenario:** Create a Mobile Banking App. So the app must be secure and handle sensitive data. What testing types should be included in the Test Plan and why?
2. **Scenario:** A software company initially developed a Student Management System using a Layered Architecture. As the number of users increased, the system became slow, difficult to maintain, and hard to scale. The development team is now considering migrating the system to a Microservices Architecture. Based on this scenario answer the following,
 - a) What are the limitations of Layered Architecture highlighted in this scenario?
 - b) How would Microservices Architecture solve these problems?
 - c) What challenges might the team face when migrating to microservices?
 - d) Which architecture would you recommend for a large-scale system and why?
 - e) Explain how scalability differs between the two architectures.

Submission

Create a **short report (PDF)** and rename it as follows:

ASD_Tutorial-<your index number>-<your name>.pdf

